

Ratio and Proportion — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong. Almost every note below comes back to one of two errors: reading $a : b$ as the fraction a/b , or dividing a total by the number of people instead of the number of parts. Watch for which of the two a given student favours — they rarely make both.

Paper A — Medium

30 MARKS • NON-CALCULATOR

- 1 Row 1 $18 : 24 \rightarrow 3 : 4$ and $3/7$ [6]
 Row 2 $20 : 50 \rightarrow 2 : 5$ and $2/7$
 Row 3 $45 : 27 \rightarrow 5 : 3$ and $5/8$
One mark per correct box. In the last column, $3/4$, $2/5$ and $5/3$ are the ratio copied out again rather than a fraction of the total — this is the single most common error in the topic. Award nothing for those. The denominator must be the sum of both parts.
- 2 (a) Both divide by 12: $2 : 3$ [2]
 (b) All divide by 5: $1 : 3 : 4$
- 3 (a) $30 \text{ cm} : 100 \text{ cm} = 3 : 10$ [2]
 (b) $45 \text{ min} : 120 \text{ min} = 3 : 8$
Answers of $30 : 1$ and $45 : 2$ score zero — the units were not converted first. This is worth pointing out every time it appears, because the student who does it once does it every time.
- 4 $2 + 5 = 7$ parts, one part = $350 \div 7 = 50$ [1]
100 and 250 [1]
- 5 $4 + 5 = 9$ parts, one part = $720 \div 9 = 80$ [1]
320 and 400 [1]
- 6 $1 + 2 + 5 = 8$ parts [1]
 One part = $960 \div 8 = 120$ [1]
₹120, ₹240 and ₹600 [1]
Dividing 960 by 3 because there are three people gives 320 each and scores nothing. The tell is that the three shares come out equal from an unequal ratio.
- 7 7 parts, one part = $28 \div 7 = 4$ [2]
 (a) **12 boys** (b) **16 girls**
Answers reversed (16 boys, 12 girls) score zero for both. The order in the question is boys to girls.
- 8 $300 \div 4 = 75 \text{ g per person}$ [1]
 $75 \times 10 = 750 \text{ g}$ [1]
- 9 $60 \div 5 = ₹12 \text{ per pen}$ [1]
 $12 \times 8 = ₹96$ [1]
- 10 (a) $60 \times 4 = 240 \text{ kg of sand}$ [2]
 (b) $60 + 240 = 300 \text{ kg}$
In (b), answering 240 means the student gave the sand again rather than the total. Award (a) only.
- 11 One part = $18 \div 2 = 9$ [2]
 (a) $3 \times 9 = 27 \text{ blue}$ (b) **45 altogether**
The 18 is two parts, not one and not the total. Students who treat it as the total get $2 : 3$ of 18 and lose both marks.

- 12** Shop A: $150 \div 2 = \text{₹}75 \text{ per kg}$ [1]
 Shop B: $365 \div 5 = \text{₹}73 \text{ per kg}$ [1]
Shop B is better value [1]

The third mark is the conclusion in words. Two correct unit prices and no statement scores 2 — this is the easiest mark on the paper to leave behind, and it is left behind constantly.

Paper B — Medium

30 MARKS • NON-CALCULATOR

- 1** (a) $250 \text{ g} : 1000 \text{ g} = 1 : 4$ [3]
 (b) $40 \text{ cm} : 150 \text{ cm} = 4 : 15$
 (c) $20 \text{ min} : 60 \text{ min} = 1 : 3$

Each mark requires the conversion. $250 : 1$, $40 : 1.5$ and $20 : 1$ all score zero.

- 2** (a) Divide both by 4: **1 : 2.5** [2]
 (b) Divide both by 5: **1 : 2.4**

A decimal on the right is correct and expected here. Students who insist on whole numbers often give $2 : 5$ for (a), which is a valid equivalent ratio but is not in the form $1 : n$ that was asked for.

- 3** (a) $6 \times 50\,000 = 300\,000 \text{ cm}$ [1]
 $300\,000 \text{ cm} = 3000 \text{ m} = \text{3 km}$ [1]
 (b) $8 \text{ km} = 800\,000 \text{ cm}$, and $800\,000 \div 50\,000 = \text{16 cm}$ [1]

In (a) the multiplication is routinely correct and the conversion routinely is not. 300 000 left in centimetres, or divided only once to give 3000, scores the first mark only.

- 4** $2.1 \div 7 = 0.3 \text{ kg per book}$ [1]
 $0.3 \times 12 = \text{3.6 kg}$ [1]

- 5** $90 \div 6 = 15 \text{ km per litre}$ [1]
 $15 \times 15 = \text{225 km}$ [1]

- 6** (a) $840 \div 105 = \text{₹}8$ [2]
 (b) $15 \times 105 = \text{₹}1575$

Multiplying in (a) or dividing in (b) means the direction was not thought about. A quick sense-check: rupees are the smaller unit, so a rupee figure is always the bigger number.

- 7** Pack A: $135 \div 900 \times 100 = \text{₹}15.00 \text{ per } 100 \text{ g}$ [1]
 Pack B: $1.2 \text{ kg} = 1200 \text{ g}$, $174 \div 1200 \times 100 = \text{₹}14.50 \text{ per } 100 \text{ g}$ [1]
Pack B is better value [1]

Pack B costs more in total and is still the better buy — that is the whole design of the question. A student who answers “A, because it is cheaper” has compared the two prices instead of the two rates.

- 8** $2 + 3 + 4 = 9 \text{ parts}$ [1]
 One part = $4500 \div 9 = 500$ [1]
1000, 1500 and 2000 [1]

- 9** 9 parts, one part = $1800 \div 9 = 200$ [1]
 Juice = $2 \times 200 = \text{400 ml}$ [1]

1400 is the water. Award the first mark only — the question named the juice.

- 10** The difference is $5 - 2 = 3 \text{ parts}$ [1]
 One part = $24 \div 3 = 8$ [1]
 Motorbikes = $2 \times 8 = \text{16}$ [1]

Treating 24 as the total gives 7 parts and nothing works out whole. The key realisation is that a difference is measured in parts just as a total is.

- 11** (a) $120 \div 8 = 15$ g per biscuit **[1]**, so $15 \times 20 = 300$ g **[1]** [3]
 (b) $180 \div 8 = 22.5$ g of flour each, and $270 \div 22.5 = 12$ biscuits **[1]**
Part (b) runs the unitary method backwards, which is the harder direction. A student who can do (a) but not (b) has learned a procedure rather than the idea.

- 12** One part = $21 \div 7 = 3$ **[1]** [2]
 Meera = $4 \times 3 = 12$ years old **[1]**
Answering 36 means the student added 4 parts to Nikhil rather than reading 21 as seven parts.

Paper C — Hard

30 MARKS • NON-CALCULATOR

- 1** The 96 blue beads are 8 parts, so one part = $96 \div 8 = 12$ **[1]** [3]
 (a) Red = $3 \times 12 = 36$ **[1]**
 (b) Total = $11 \times 12 = 132$ **[1]**
Dividing 96 by 11, or treating 96 as the total, are the two failure modes. Ask: how many parts is the number I have been given?
- 2** The difference is $9 - 5 = 4$ parts **[1]** [3]
 One part = $36 \div 4 = 9$ **[1]**
45 and 81 [1]
Check: $81 - 45 = 36$. Dividing 36 by 14 (the sum) is the standard wrong move — that would be right only if 36 were the total.
- 3** The largest share is 7 parts, so one part = $840 \div 7 = 120$ **[1]** [3]
 Total = 15 parts **[1]**
 $15 \times 120 = ₹1800$ **[1]**
- 4** (a) Make B match at 12: A : B = 2 : 3 becomes 8 : 12 ($\times 4$), and B : C = 4 : 5 becomes 12 : 15 ($\times 3$) **[1]** [4]
 So A : B : C = **8 : 12 : 15 [1]**
 (b) $8 + 12 + 15 = 35$ parts, one part = $105 \div 35 = 3$ **[1]**
24, 36 and 45 [1]
Answering 2 : 3 : 5 in (a) scores zero and usually costs (b) as well. It is worth showing the student that 2 : 3 : 5 on 105 gives 21, 31.5 and 52.5 — half a marble twice over, so the method refutes itself.
- 5** Students appear as 18 in one ratio and 3 in the other; scale S : C by 6 **[1]** [3]
 S : C = 3 : 1 becomes 18 : 6, so T : S : C = 1 : 18 : 6 **[1]**
 T : C = **1 : 6 [1]**
The same joining move as question 4, but the shared quantity sits in the middle of one ratio and the start of the other. Award full marks for any correct scaling route.
- 6** The job is $6 \times 10 = 60$ worker-days **[1]** [3]
 (a) $60 \div 4 = 15$ days **[1]**
 (b) $60 \div 5 = 12$ workers **[1]**
The direct-proportion answer to (a) is $4/6 \times 10 = 6.67$ days, and it should be rejected on sight: fewer workers cannot finish sooner. Award nothing for it, but do point at the direction rather than the arithmetic.
- 7** Let one part be k, so Anil has 5k and Bina has 3k **[1]** [4]
 After the transfer: $5k - 120 = 3k + 120$ **[1]**
 $2k = 240$, so $k = 120$ **[1]**
 Total = $8k = ₹960$ **[1]**
Check: Anil 600, Bina 360, and after the transfer both hold 480. A common wrong route is to treat the ₹120 as one part, which gives 960 by accident on this question — ask to see the equation before awarding full marks.

- 8** The job is $4 \times 9 = 36$ pipe-hours **[1]** [3]
 (a) $36 \div 6 = 6$ hours **[1]**
 (b) More pipes must mean less time, but the student's method makes the time **longer** — he has used direct proportion where the quantities are inversely proportional **[1]**
Accept any answer in (b) that identifies the direction as wrong. Naming it "inverse proportion" is not required, but describing the multiplication as constant is a sign the idea has landed.
- 9** $5 + 2 = 7$ parts, one part = $21 \div 7 = 3$ kg **[1]** [4]
 Rice 15 kg and lentils 6 kg **[1]**
 $15 \times 60 = 900$ and $6 \times 95 = 570$ **[1]**
 Total = **₹1470 [1]**
Two ideas stacked: share in a ratio, then cost each part separately. Averaging the two prices to ₹77.50 per kg and multiplying by 21 gives ₹1627.50 and scores nothing — the two quantities are not equal, so their prices cannot be averaged.

Paper D — Hard

30 MARKS • NON-CALCULATOR

- 1** Let one part be k : boys = $7k$, girls = $5k$ **[1]** [4]
 After six more girls: $7k : (5k + 6) = 7 : 6$, so $6 \times 7k = 7(5k + 6)$ **[1]**
 $42k = 35k + 42$, so $7k = 42$ and $k = 6$ **[1]**
 Boys = $7 \times 6 = 42$ **[1]**
Check: girls go from 30 to 36, and $42 : 36 = 7 : 6$. The number of boys is unchanged throughout, which is why the same k can be used in both ratios — that is the whole idea of the question.
- 2** Let the numbers be $3k$ and $5k$ **[1]** [4]
 $(3k + 4) : (5k + 4) = 2 : 3$, so $3(3k + 4) = 2(5k + 4)$ **[1]**
 $9k + 12 = 10k + 8$, so $k = 4$ **[1]**
 The numbers are **12 and 20 [1]**
Check: $16 : 24 = 2 : 3$. Adding 4 to each side of the ratio itself, giving $7 : 9$, is the trap — you add to the numbers, never to the parts.
- 3** The order is $12 \times 15 = 180$ machine-days **[1]** [4]
 (a) $180 \div 9 = 20$ days **[1]**
 (b) $180 \div 10 = 18$ machines **[1]**
 (c) The total work is fixed, so the machines and the days multiply to a constant — increasing one must decrease the other **[1]**
For (c), accept any answer identifying the product as constant. "Because it is inverse proportion" on its own is naming rather than explaining; push for the sentence about the fixed amount of work.
- 4** Distance = $60 \times 5 = 300$ km **[1]** [3]
 (a) $300 \div 75 = 4$ hours **[1]**
 (b) $300 \div 3 = 100$ km/h **[1]**
The constant here has a physical name the student already knows — the distance. Pointing that out links this straight back to the speed topic.
- 5** (a) $6 : 9 : 12$ simplifies to $2 : 3 : 4$ **[1]** [4]
 9 parts, one part = $4500 \div 9 = 500$ **[1]**
 Asha **₹1000**, Bhavna **₹1500**, Chetan **₹2000 [1]**
 (b) Chetan is entitled to $4/9$ of the profit, not $1/2$. Half would be ₹2250, so he is claiming **₹250 too much [1]**
Working directly with $6 : 9 : 12$ without simplifying is perfectly valid and reaches the same shares — do not penalise it. In (b), $4/9 < 1/2$ is enough on its own; the money comparison is the fuller answer.

- 6** (a) Shop A: $99 \div 1.5 = \text{₹}66 \text{ per litre}$ [1] [4]
 Shop B: $126 \div 2 = \text{₹}63 \text{ per litre}$, so **Shop B is better value** [1]
 (b) Three bottles cost $2 \times 99 = \text{₹}198$ and give 4.5 litres [1]
 $198 \div 4.5 = \text{₹}44 \text{ per litre}$, so **Shop A is now better value** [1]
The offer changes the answer completely, which is the point. The frequent error in (b) is dividing the price of three bottles by three bottles' worth of litres while forgetting that only two were paid for.
- 7** Flour: $500 \div 200 \times 12 = 30$ pancakes [1] [3]
 Milk: $700 \div 300 \times 12 = 28$ pancakes · Eggs: $6 \div 2 \times 12 = 36$ pancakes [1]
 The smallest wins: **28 pancakes, limited by the milk** [1]
The third mark needs both the number and the ingredient. Adding or averaging the three figures scores nothing — you cannot make a pancake out of the ingredient you have run out of.
- 8** (i) The mistake: reading the colon as a fraction bar. There are $3 + 5 = 8$ parts, not 5 [4]
[1]
 The correct fraction is **$\frac{3}{8}$** [1]
 (ii) The mistake: B is 3 parts in one ratio and 4 in the other, so the two cannot be written side by side [1]
 Scaling both until B is 12 gives **$A : B : C = 8 : 12 : 15$** [1]
These are the two errors the whole topic is built around, which is why they close the paper. A student who can diagnose both in someone else's handwriting will not make them in their own — that is the entire reason this question type exists.